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Simplifying the Medicare Plan Finder tool could help older adults choose lower cost Part D plans

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Abstract

Helping older adults make good plan choices is a persistent challenge of the Medicare Prescription Drug (Part D) Program. CMS currently provides an internet-based decision support tool (Plan Finder), but it appears to have had limited effects due, in part, to the tool being complex and difficult to interpret. This study uses a randomized experiment with hypothetical Part D plan choices to test the effect of simplifying the default amount of financial information provided on Plan Finder on individuals' ability to select low cost plans. Reducing the amount of financial information displayed results in the selection of lower cost plans, with no accompanying decrease in average plan quality or pharmacy network size but an increase in the take-up of convenience options like a mail-order pharmacy. These modifications to the current Plan Finder design have the potential to improve the tool's usability and beneficiaries' plan choices in the Part D market.

Background

The Medicare Prescription Drug Program (Part D) has been successful along several dimensions. It significantly increased Medicare beneficiaries' access to prescription drug coverage,(1) lowered older adults' out-of-pocket spending on prescription drugs,(2) and has incurred lower-than-expected program costs.(3, 4) A persistent challenge facing this program, however, has been Medicare beneficiaries' ability to shop for and choose Part D plans that best match their needs, budgets, and preferences.

A sizable body of evidence indicates that suboptimal plan choices are pervasive. The vast majority of Part D enrollees do not choose their lowest cost plan.(5) After accounting for the amount of protection plans provide against unanticipated drug costs, individuals are still estimated to overspend by about 30% of their total drug spending each year.(6, 7) Beneficiaries rarely switch plans, despite annual changes to plans' costs and benefits,(8, 9) and it appears that plan selection has gotten worse over time despite older adults gaining more experience with the Part D program.(7) Furthermore, surveys of beneficiaries reveal that a large majority find the Part D program too complicated,(10) with many citing the number of plans and difficulty understanding specific plans' formularies as key sources of this complexity.(11) Evidence indicates that when the decision task is simplified through the provision of a letter containing personalized plan recommendations, Part D enrollees are more likely to switch to lower cost plans.(12) Combined with evidence from health insurance markets that plan choices improve as the decision environment is simplified through plan standardization,(13) these findings suggest that many older adults may be overwhelmed by the complexity of selecting a Part D plan.

The Plan Finder decision support tool is one of the primary mechanisms used by the Centers for Medicare and Medicaid Services (CMS) to assist beneficiaries with identifying a preferred plan from among the roughly 40 options available on average(14, 15). Plan Finder's interactive website provides enrollees with a complete list of Part D plans available in their ZIP code, as well as plan details like premiums, cost-sharing amounts, five-star quality scores, and formulary design. After users enter their current prescription drug regimen, the tool also provides personalized total cost estimates for each plan based on individuals' current drug use. By default, Plan Finder sorts available plans by this total cost estimate, listing the financial and non-financial details for the lowest-cost plan first. The combination of default plan sorting and the presentation of personalized total cost estimates should, in theory, "nudge" individuals towards choosing the plan that minimizes their total drug costs. Yet, the evidence described above indicates that this tool has not been widely effective.

Some of this ineffectiveness is likely due to the apparently limited use of the tool. A 2014 Government Accountability Office report indicated that in 2013 only 14% of stand-alone Part D plan enrollments were completed through the Plan Finder website.(16) Evidence also exists, however, that individuals who attempt to use Plan Finder find the tool difficult to use. For example, users report that the tool is confusing and overly technical,(17) and they have difficulty navigating the website and identifying pertinent information.(18) As a result, changes to Plan Finder that simplify the amount of information presented and increase the emphasis on personalized total cost estimates may improve the effectiveness of this tool in promoting the selection of low cost prescription drug plans.

CMS has previously expressed interest in modifying the design of Plan Finder, including reducing the amount of plan information that is shown by default.(19) Earlier studies of such simplifications for prescription drug and health insurance plan menus have shown encouraging results,(19–21) yet important questions remain unanswered. First, it is unclear how much or how little information should be displayed. Prior research in this area has focused on testing the effect of displaying total cost estimates alongside plan premiums. However, other approaches to simplification may produce different choice outcomes. Second, the effect of Plan Finder changes are unknown when the menu of available plans does not contain a clearly dominant option (i.e., a plan that is objectively as good or better than all other plans in the choice set across all plan characteristics). Because real-world Part D choices require trade-offs between competing plan attributes, this lack of information represents an important limitation of the current literature in this area. Third, it is unclear how modifying the amount of financial information on the default plan menu influences choices when full information can be obtained through additional research or searching. Plan Finder is interactive, meaning that users can access additional plan details by clicking available links within the menu. As a result, any simplification to the default menu would likely be accompanied by an option to view all financial plan characteristics, potentially altering the effects previously studied changes to existing decision support tools.

This study uses a randomized experiment to test the effect of several Plan Finder simplifications on individuals' Part D plan choices and their perceptions of the decision support tool and the information it provides. It addresses the aforementioned knowledge

gaps in several ways. First, it tests several different forms of information simplification to better understand how certain Plan Finder changes are likely to influence plan choices. Second, it presents respondents with plan menus that simulate the actual choices faced on the Part D market, without a clearly dominant option, and examines how default menus influence trade-offs between particular plan attributes. Finally, it uses interactive plan menus that allow users to “opt-out” of the intervention and see full plan information upon request, to help provide realistic estimates of anticipated treatment effects in a field setting.

Methods

Data Source

This experiment was administered through the American Life Panel, a nationally representative internet-based panel recruited and maintained by the RAND Corporation. 1,785 panel members aged 55 and older were recruited to participate in the survey and 1,321 agreed (a completion rate(22) of 74%). We further restrict the study sample to 1,278 study participants who completed at least one of the choice tasks. Individuals under the Medicare eligibility age (65) were intentionally included in this study for two reasons. First, qualitative evidence suggests that older adults frequently get help from younger individuals, such as adult children, when making Medicare decisions. As a result, Plan Finder users likely include individuals younger than 65. Second, it is important to understand how changes to the Plan Finder tool will impact new cohorts of Medicare beneficiaries who will age into the program in the coming years. The inclusion of individuals who are near Medicare eligibility facilitates such analyses.

Experimental Design

All subjects were asked to make a hypothetical Part D plan choice on behalf of a 65-year old friend, Mrs. Smith. Participants were given a vignette prior to making plan selections which described Mrs. Smith’s current prescription drug use and her preference for minimizing her total prescription drug costs for the year. The vignette also highlighted Mrs. Smith’s secondary preferences, namely having a plan with a good quality rating and having options over where to fill her prescriptions. The full text of the vignette is available in the technical appendix.(23) Respondents are asked to choose a plan for someone other than themselves to ensure that all participants are making plan choices based on a similar set of preferences (i.e., minimize total costs while giving secondary priority to plan quality and options over where prescriptions can be filled). This restriction allows for the testing of whether Plan Finder simplifications can improve users’ ability to choose plans that are consistent with an intended decision rule and enables normative inferences regarding whether the quality of plan choices improved across study arms.

Respondents were then presented with a menu of 10 randomly selected plans that mimic actual plans available in the 2017 Part D market, thereby simulating the trade-offs individuals must navigate when making actual plan choices. Refer to the technical appendix for additional information on how simulated plans were generated.(23) The information displayed on the plan menu varied across the four study arms (one control and three treatment) to which participants were randomly assigned.

Control Group

The control group saw a menu that mirrors the current Plan Finder design by displaying a total cost estimate (obtained by applying each plan's formulary and financial attributes to one's current drug regimen), premiums, deductibles, copayment ranges, and non-financial plan features (i.e., five-star quality rankings, number of in-network pharmacies, and the availability of a mail-order pharmacy) for each plan. A screenshot of this status-quo menu is provided in the technical appendix.(23)

Treatment Groups

The financial information displayed on the default menu was simplified to varying degrees across the three experimental study arms. Simplifications were aimed at increasing the emphasis on the personalized cost estimates and reducing the amount of information concerning the specific cost-sharing details used to calculate these estimates for each plan (i.e., deductibles and copayment ranges). All treatment groups had the option to override the default menu and view the full financial information available to the control group by clicking on a link within the plan menu. Non-financial information was displayed in a manner identical to the control group. Descriptions of the default presentation format for the treatment groups are below and screenshots of the plan menus are in the technical appendix. (23)

Total Cost Only: Subjects in this study arm saw only the personalized annual total cost estimate alongside non-financial plan attributes.

Total Cost and Premium: Monthly premiums were shown by default in addition to the total cost estimate and non-financial plan characteristics.

Total Cost, Premium, and Out-of-Pocket Cost: This arm was shown the total cost estimate plus its two primary components: monthly plan premiums and estimated out-of-pocket costs (i.e., total costs less annual premiums).

Additional Study Details

Respondents in each treatment arm completed the decision task three times, with a new set of available plans being randomly generated each iteration. Repeating the decision tasks increased the study's effective sample size and ensured that respondents encountered diverse choice sets with different types of trade-offs between plans. Respondents also answered a series of questions about their demographics, current health status, prescription drug use, and numeracy (i.e., their ability to work with and understand numbers). Details of the numeracy scale used in this study are available in the technical appendix.(23) Finally, study participants were asked to describe their perception of the decision support tool and their plan choices. Specifically, they were asked whether they felt they had too much, not enough, or about the right amount of information, and how confident were they that the plan chosen best met Mrs. Smith's needs and preferences.

Analytic Approach

The primary outcome of interest in this study was the total cost of the selected plan relative to the total cost of the other plans available in a given choice set. We examined the effect of

Plan Finder simplifications by evaluating whether the cumulative distribution of plans chosen by total cost rank (i.e., the cheapest plan in the choice set has a rank of 1, while the most expensive plan has a rank of 10) differed between the control group and the three treatment groups. We also calculated average “excess spending” (i.e., the difference between the estimated total cost of the plan chosen and the lowest total cost plan in the choice set), and tested for differences in the average value of this outcome. It should be noted that, at an individual-level, choosing a plan with expected costs larger than the lowest cost plan do not necessarily reflect a suboptimal decision because these added costs may be “buying” other important plan features, like a higher quality score or additional network pharmacies. As a result, we also examine average values of the non-financial characteristics of the selected plan, between the treatment and control groups to assess whether changes in cost-sensitivity result in inattention to other plan features. Estimates were clustered at the respondent level to account for the repeated decision tasks. Differences in reported satisfaction with the amount of information provided by the tool and confidence in the plans selected were also examined between the control group and three treatment groups. In subgroup analyses, we examined whether the treatment effects varied by respondent characteristics, including age, gender, education, numeracy, family income, and self-reported health status.

Limitations

This study had several limitations. First, our experiment relied on hypothetical choices without real-word consequences. As a result, a natural concern is that survey respondents may not approach the decision tasks in the same manner they would when facing actual financial incentives. However, engagement with the decision tasks appeared to be quite high. Over 75% of the study sample found the survey to be interesting or very interesting, and plan choices within the control group resembled actual Part D choices in a number of ways. For instance, respondents tended to choose plans with lower premiums and lower deductibles, and they preferred plans with higher quality rankings, all else being equal.⁽⁷⁾ As such, it appears that study participants took seriously the task of choosing a Part D plan.

Second, respondents were asked to make a selection on behalf of someone else (“Mrs. Smith”) rather than themselves. This limits the ability of our findings to speak to whether individuals with differing preferences and anticipated drug needs would vary in their responses to changes in Plan Finder’s default menu. However, this design feature has several advantages. It created consistency in the decision task across respondents, eliminated issues around brand (i.e., insurer) preferences or loyalty, and enabled the direct evaluation of whether Plan Finder simplifications improve individuals’ ability to identify and select cost-minimizing Part D plans. Future research should examine to what extent older adults prefer cost-minimizing Part D plans and how the design of Plan Finder’s default menu influences enrollment decisions for individuals with different priorities for their drug coverage.

A final limitation of this study is related to the generalizability of our results to the older adult population. In particular, it is possible that the individuals who were willing to participate in this survey differed from the general population. The America Life Panel has two important features that helped to mitigate this potential source of selection bias. First, it used nationally representative probability sampling to recruit survey participants. Second, a

computer and internet access were provided by RAND for sampled individuals without access to this technology, ensuring that prior internet use was not a prerequisite to study participation. These features improve generalizability and represent a significant improvement over studies conducted with convenience samples. In addition, we conducted subgroup analyses that focus on those 65 and older and populations who may be underrepresented in our overall study sample (e.g., individuals with lower income or educational attainment). These examinations provided insight into the extent to which overall treatment effects vary by particular respondent characteristics and whether they are likely to generalize to Plan Finder users at large.

Results

The sample was predominantly white and had good self-reported health (Exhibit 1). About half of the sample had at least some college education, and slightly less than half had family incomes above \$75,000 a year. Fifty-one percent of the sample was a current Medicare beneficiary, and about 60% of these individuals were enrolled in Part D, similar to national estimates of Part D participation.(24) Numeracy scores were high relative to other samples of older adults.(25) In additional analyses, we tested whether participants were balanced on these and other observable characteristics across the four study arms; refer to the technical appendix for details.(23) Results indicated good balance, consistent with an effective randomization process.

When examining the cumulative proportion of plans chosen by their total cost rank across the four study arms, we found that individuals in the Total Cost Only and the Total Cost, Premium, and Out-of-Pocket Cost groups selected the lowest or second lowest cost plan about 65% of the time (Exhibit 3). In comparison, individuals in the Total Cost and Premium group selected such plans 54% of the time, while control group participants did so just 44% of the time. Overall, the shape of the cumulative distributions clearly indicates that the treatment groups were more likely to choose lower-cost plans and less likely to choose higher-cost plans than the control group.

Consistent with the distributional shifts noted in Exhibit 2, we found that simplifying the financial information on Plan Finder resulted in significantly lower excess spending (Exhibit 3). On average respondents in the control group selected plans that cost \$411 more than the cheapest plan, while the respondents in the Total Cost Only, Total Cost and Premium, and Total Cost, Premium, and Out-of-Pocket groups had average excess spending amounts of \$209, \$307, and \$192, respectively. In other words, overspending was 49% lower in Total Cost Only group ($P<0.001$), 25% lower in the Total Cost and Premium group ($P<0.001$), and 53% lower in the Total Cost, Premium, and Out-of-Pocket group ($P<0.001$) relative to controls.

With respect to non-financial plan features, Plan Finder modifications had no effect on the average quality of the chosen plans (Exhibit 3). We find respondents in all three treatment groups were more likely on average to choose a plan with a mail order option relative to the control group (consistent with Mrs. Smith's secondary preferences). We also found that the Total Cost, Premium, Out-of-Pocket group chose plans with a statistically significantly

lower number of network pharmacies in Mrs. Smith's ZIP code, although the difference was small in magnitude (3.14 vs. 3.25 pharmacies; $P = 0.019$).

Information reductions appeared to have limited effects on respondents' confidence in their choices. All treatment groups had modestly higher rates of being confident that they chose the best plan to meet Mrs. Smith's needs, but this increase was only significant at a 90%-level in the Total Cost and Premium group ($P = 0.095$). Simplifications had no significant effects on the number of users who felt they were given the right amount of information. The Total Cost Only (19.3%) and Total Cost and Premium (16.1%) groups were less likely to report having too much information ($P = 0.0136$ and $P < 0.001$, respectively); however most of these improvements were offset by increases in the number reporting they did not have enough information (17.4% vs. 26.3% in the Total Cost Only group [$P = 0.005$] and 22.6 in the Total Cost and Premium group [$P = 0.102$]).

We found little evidence that the overall effects of Plan Finder simplifications on excess spending differed across particular subgroups (Exhibit 4). Reductions in excess spending across treatment groups did not statistically differ by education, numeracy, income, and self-reported health status. An exception to this pattern occurred with respect to gender, where we found that females experienced greater reductions in excess spending, relative to males, in all treatment groups. Similarly, older adults in the treatment groups appeared to experience larger reductions compared to respondents under the age of sixty-five, though these differences did not reach statistical significance at a 90% confidence level. Refer to the technical appendix for additional details on estimated treatment effects.(23)

Discussion

The Part D program relies on consumer choice to promote competition between private insurers and ensure a good fit between beneficiaries' prescription drug coverage and their drug needs and preferences. As a result, it is critical that older adults have the resources they need to effectively evaluate their prescription drug coverage options and make good plan choices. Although the Plan Finder tool was intended to be this resource, recent evidence suggests that it has not served Medicare beneficiaries well in its current form. This study demonstrates that simplifying the financial information presented by default on Plan Finder may be one straightforward approach to improving the tool's usability and facilitating better comparisons across plans on the basis of total expected cost.

All of the presentation formats tested in this study improved users' ability to select cost-minimizing plans compared with the way financial information is currently presented on the Plan Finder website. We find that these effects are largely consistent across a number of individual characteristics, with one statistically significant exception. Women in the three treatment groups experienced larger reductions in excess spending relative to men in the same study arm.

Differences in the degree of improvement were noted across the treatment arms. Specifically, the excess spending reductions were roughly twice as large when only total cost estimates were shown or when total cost was presented alongside plan premiums and an

estimate of annual out-of-pocket costs. Further research is needed to understand precisely why improvements were smaller within the Total Cost and Premium group, but a potential explanation is that users in this study arm had difficulty determining how total cost estimates and premiums relate to each other. For example, users may have interpreted premiums as costs *in addition* to total costs rather than a component of total costs (despite the availability of definitions that detailed how total costs were calculated). Alternatively, users may have simply focused more on plan premiums instead of total costs when choosing a plan for Mrs. Smith. These potential valuation errors are addressed in the other two treatment arms through notably different approaches. In the Total Cost Only group, premiums are hidden by default, thereby decreasing the chance for erroneous calculations and over-emphasis on plan premiums compared to total costs. In the Total Cost, Premium, and Out-of-Pocket Cost group, both components of the total cost estimate are displayed side-by-side. This increased transparency may allow users to better visualize how total costs are calculated and facilitate equal emphasis on expected costs arising from premiums and out-of-pocket spending.

Users appear to be receptive to the simplifications tested in the current study. A majority of respondents reported they had about the right amount of plan information, regardless of study arm. Additionally, the increased emphasis on personalized cost estimates did not appear to have the unintended consequence of causing users to focus solely on total cost estimates and neglect non-financial plan characteristics like plan quality. In fact, reducing the amount of financial information appears to have allowed all treatment arms to more frequently choose plans with a mail order option, which increases one's choice over where to get prescriptions filled. This suggests that simplifying financial information through the use of summary measures of cost may make it cognitively easier for users to incorporate additional plan features into their decision-making process.

It should be noted that this study does not evaluate the utility of making plan selections according to personalized total cost estimates that are based on the prescription drugs an individual is currently taking. Previous research has demonstrated that current drug use can produce underestimates of actual drug costs for the coming year,(26) potentially leading to inadequate risk protection from unanticipated drug needs. Future research should examine how best to summarize the level of risk protection particular Part D plans provide and how to help Medicare beneficiaries incorporate this information into their decision making process. Nevertheless, the current design of the Medicare Plan finder tool implicitly encourages beneficiaries to choose plans based on the costs of current drug use, and in the absence of additional information, current drug use remains a good predictor of actual drug use in the upcoming year.(26) As a result, choices based on the current total drug cost estimate likely create a good starting point for annual plan selections.

Conclusion

Simplifying the Plan Finder menu by displaying total costs alone or total costs, premium, and out-of-pocket estimates in lieu of complicated cost-sharing details results in the selection of lower cost plans, with no accompanying decrease in average plan quality or pharmacy network size but an increase in the take-up of convenience options like a mail-

order pharmacy. These changes may be a low-cost policy approach to facilitating improved consumer decision making within the Part D market.

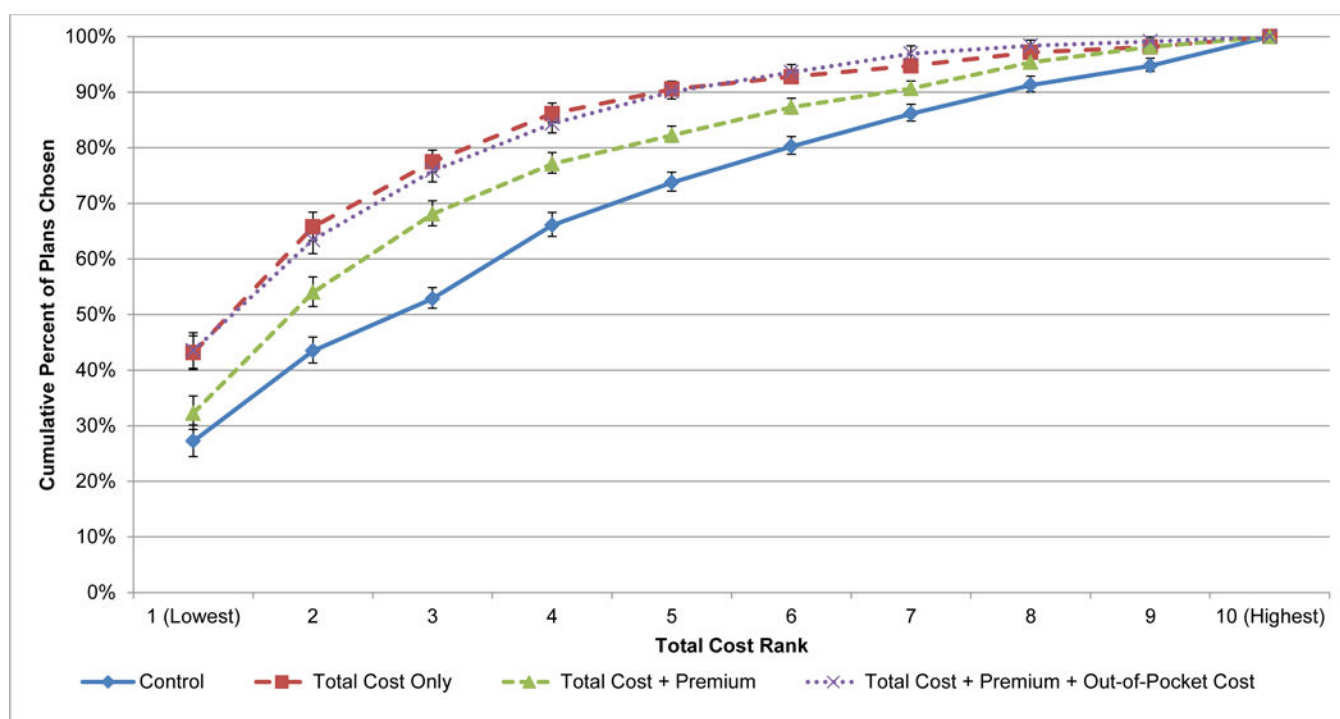
Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Source: Authors' calculations using the 2017 online survey conducted with the American Life Panel.

Notes: Total cost rank is calculated by determining where a chosen plan ranks, in terms of the total cost estimate, relative to the other plans in a given choice set. A rank of 1 means the selected plan was the cheapest plan in the choice set. Error bars represent 95%-confidence intervals of proportion estimates, accounting for clustering at the respondent level.

Exhibit 2.

“Cumulative distribution of selected plans according to total cost rank by treatment arm”

Exhibit 1-

Sample characteristics (n=1,278)

Race	White, non-Hispanic (%)	82.7
Gender	Female (%)	50.2
Married	(%)	57.7
Education	At least some college (%)	51.0
Family Income	\$75k or more (%)	46.3
Medicare beneficiary	(%)	51.3
Enrolled in Part D	(%)	31.1
Age	(mean; SD)	66.2 (7.7)
Self-reported health	Fair-Poor (%)	16.2
Number of drugs currently taking	(mean; SD)	3.5 (3.4)
Numeracy score (0–3)	(mean; SD)	2.3 (0.9)
Study Arm	Control (%)	25.0
	Total cost only (%)	26.8
	Total cost and premium (%)	24.3
	Total cost, premium, and Out-of-Pocket costs (%)	24.0

Source: Authors' calculations using the 2017 online survey conducted with the American Life Panel. **Notes:** SD = standard deviation. Estimates are unweighted.

Exhibit 3-

Summary of the effects of Plan Finder simplifications

	Control	Total Cost Only	Total Cost and Premium	Total Cost, Premium, and Out-of-Pocket Cost
Excess spending (mean, \$)	411.06	208.78 ****	306.94 ***	191.63 ****
Plan quality (mean, 1–5 stars)	3.51	3.50	3.51	3.53
Number of in-network pharmacies (mean, 1–5)	3.25	3.26	3.23	3.14 **
Mail-order available (%)	72.83	80.99 ****	76.45 *	77.31 **
Amount of plan information (%)				
Too much	27.44	19.3 **	16.13 ****	24.01
Not enough	17.35	26.32 ****	22.58	17.43
About right	55.21	54.39	61.29	58.55
Confidence that chosen plan was best (%)	51.42	56.73	58.06 *	55.59

Source: Authors' calculations using the 2017 online survey conducted with the American Life Panel. **Notes:** Statistical tests evaluate differences in the estimated values for the respective treatment groups relative to the control group. Estimates for excess spending, plan quality number of network pharmacies, and the availability of mail order options are clustered at the respondent level to account for repeated observations.

* Indicates statistical significance at the $p = 0.1$ level.

** Indicates statistical significance at the $p = 0.05$ level.

*** Indicates statistical significance at the $p = 0.01$ level.

**** Indicates statistical significance at the $p < 0.001$ level

Exhibit 4-

The effects of Plan Finder simplifications on excess spending by respondent characteristics

		Average Excess Spending by Respondent Characteristics (\$)	Difference in Excess Spending from Control Group (\$)		
		Control Group	Total Cost Only	Total Cost and Premium	Total Cost, Premium, and Out-of-Pocket Cost
Age	55–64	358.46	–158.33	–64.35	–172.81
	65+	458.22	–241.39	–140.72	–261.71
Gender	Male	331.72	–141.89	–21.38	–167.12
	Female	478.71	–249.36 *	–174.99 **	–257.65 *
Education	High school or less	454.11	–242.80	–126.89	–218.67
	At least some college	368.90	–162.40	–83.59	–216.99
Numeracy Score	0–2	478.05	–226.50	–136.72	–235.64
	3	348.96	–182.20	–71.46	–215.92
Family Income	Less than \$75,000	456.42	–214.46	–146.79	–246.40
	\$75,000 or more	364.65	–194.97	–61.25	–192.41
Health Status	Poor - fair	474.17	–226.84	–65.52	–210.67
	Good - Excellent	397.60	–196.21	–110.22	–218.64

Source: Authors' calculations using the 2017 online survey conducted with the American Life Panel. **Notes:** Statistical tests evaluate whether the change in excess spending relative to the control group differs by the respondent characteristic of interest. Estimates are clustered at the respondent level to account for repeated observations.

* Indicates statistical significance at the $p = 0.1$ level

** Indicates statistical significance at the $p = 0.05$ level.

*** Indicates statistical significance at the $p = 0.01$ level.

**** Indicates statistical significance at the $p < 0.001$ level.